INCH-POUND

MIL-DTL-43741C
June 6, 2007
SUPERSEDING
MIL-N-43741B
30 August 1989

DETAIL SPECIFICATION

NECKERCHIEF, COTTON, KNITTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two classes of neckerchiefs made from cotton knitted cloth.

1.2 Classification. The neckerchief is furnished in the following classes

Class 1 - Brown 436
Class 2 - Foliage Green 504

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Section 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: Mary.Lockley@dla.mil, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database at http://assist.daps.dla.mil/.

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COMMERCIAL ITEM DESCRIPTION

A-A-50199 - Thread, Polyester Core, Cotton or Polyester-Covered

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/, or http://assist.daps.dla.mil, or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

FEDERAL ACQUISITION REGULATIONS (FAR)

52.209-4 - First Article Approved – Government Testing

(Copies are available online at http://acquisition.gov/far/index.html or by contacting the Superintendent of Documents at 202-512-1800.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Test Method 8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC Test Method 15 - Colorfastness to Perspiration
AATCC Test Method 16 - Colorfastness to Light
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 20A - Fiber Analysis: Quantitative
AATCC Test Method 61 - Colorfastness to laundering, Home and Commercial: Accelerated
AATCC Test Method 81 - pH of the Water-Extract from Wet-Processed Textiles
AATCC Evaluation Procedure 1 - Gray Scale for Color Change
AATCC Evaluation Procedure 2 - Gray Scale for Staining
AATCC Evaluation Procedure 8 - AATCC 9-Step Chromatic Transference Scale
AATCC Evaluation Procedure 9, Option A - Visual Assessment of Color Differences of Textiles

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(Copies are available online at www.aatcc.org, or from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies are available online at http://www.asq.org, or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

ASTM INTERNATIONAL

ASTM D 276 - Standard Test Method for Identification of Fibers in Textiles
ASTM D 629 - Standard Test Methods for Quantitative Analysis of Textiles
ASTM D 1776 – Standard Test Method for Conditioning and Testing Textiles
ASTM D 3776 - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D 3787 - Test Method for Bursting Strength of Textiles – Constant-Rate-of Traverse (CRT) Ball Burst Test
ASTM D 3887 - Standard Specification for Tolerances for Knitted Fabrics
ASTM D 6193 - Standard Practice for Stitches and Seams

(Copies of documents are available online at www.astm.org, or from the ASTM INTERNATIONAL, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

OTHER PUBLICATIONS

Repeat Insult Patch Test - Modified Draize Procedure

Principles and Methods of Toxicology (fourth edition), A Wallace Hayes (editor), pp 1057 - 1060, 2001.

(Copies are available at http://www.taylorandfrancis.co.uk/ or from Taylor and Francis, 325 Chestnut Street, Philadelphia, PA 19106.)

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 and 6.3.

3.2 Standard sample. The finished Neckerchief shall match the standard sample for shade and appearance, and shall, unless otherwise indicated, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Cloth. The cloth for the neckerchief shall be knitted (see 6.5) from cotton yarns. The cloth shall meet the requirements in Table I when tested as specified in 4.3.4.

TABLE I. Cloth requirements

Characteristics	Requirement
Fiber identification	100% cotton
Knit type	Circular knit - flat jersey
Weight, oz/sq. yd (min.)	2.5
Fabric count, per inch (min.) Wales Courses	 28 25
Bursting strength, lbs. (min.)	35
Non-fibrous content, % (max.)	6.0
pH	5.0-8.5

3.5 Color. The color of the knitted cloth shall be Brown 436 for Class 1 and Foliage Green 504 for Class 2. Dyes or compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used

3.5.1 Labile Sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used

3.5.2 Visual shade matching (both Classes). The color and appearance of the cloth shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 ± 200K illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 ± 200K.

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3.5.3 Colorfastness. The dyed and finished cloth shall conform to the colorfastness requirements listed in Table II when tested as specified in 4.3.4.

TABLE II. Colorfastness requirements (all Classes)

Color Evaluation	Laundering (3 cycles) 1/ (min.)	Light (40 hrs or 170KJ) 2/ (min.)	Perspiration 1/ Acid and Alkaline (min.)	Crocking 3/ (min.)
Brown 436	3-4	3-4	3-4	3.5
Foliage Green 504	3-4	3-4	3-4	3.5

1/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
2/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change.
3/ Rated using the AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale.

3.6 Toxicity. The finished neckerchief shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.4.1. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.7 Thread. The thread for over-edge stitching shall be Cotton Covered Polyester Core, Type I, Tex 31-35, 2 ply of A-A-50199, dyed to match Brown 436 for Class 1, and Foliage Green 504 for Class 2.

3.8 Label. The identification label shall conform to Type III or IV, Class 1 of MIL-DTL-32075. The label shall show colorfastness to laundering as specified in MIL-DTL-32075. The label shall be positioned on the back of the neckerchief, 1 ± 1/8 inches from one of the four corners.

3.8.1 Bar Code Labels. Shall be as specified in the contract (see 6.2).

3.9 Design and size. The neckerchief shall be a single layer of the knitted cloth, with raw edges over-edge stitched. Finished dimensions shall be 72 ± 1 inches long by 24 ± 1 inches wide.

3.10 Construction.

3.10.1 Manufacturing. The neckerchiefs shall be manufactured utilizing the contractor's best commercial practices.

3.10.2 Stitches and seams Stitches and seam types specified in Table III shall conform to ASTM D 6193. When two or more stitch types are given for the same part of the operation, any one of them may be used.

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TABLE III. Stitches and seams

	Stitch type	Seam type	Thread		
			Needle	Bobbin/ looper	Cover
Over-edge stitching. Finished appearance. When a knife is used on an over-edging machine, care shall be exercised that the edge of the fabric is not cut off but the raw edge is trimmed. Over-edge stitch the four raw edges of neckerchief. The gauge of stitching shall be 1/8 to 3/16 inch.	502 or 503	EFd-1	70-3	70-3	
The ends of stitching shall overlap 3/4 to 1-1/4 inches.	504 or 505	EFd-1	70-3	70-3	70-3

3.11.2.1 Stitches per inch. The seams shall be sewn with 6 to 10 stitches per inch.

3.11.3 Thread breaks. Thread breaks shall be securely stitched back of the break not less than ½ inches.

3.12 Workmanship. The finished neckerchief shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels (see 4.3).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance 3.1, shall consist of examinations and tests as specified in 4.3.2, 4.3.3 and 4.3.4.

4.2.1 First article samples. Unless otherwise specified in the procurement document, first article samples shall be provided. The sample size will be specified in the procurement document. The sample shall be one neckerchief and the lot size shall be expressed in units of neckerchiefs.

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4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.3.2 and 4.3.3 and the tests of 4.3.4. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4 and with quality acceptance limits as specified in the contract and/or order, except where otherwise indicated.

4.3.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents.

4.3.2 End item visual examination. The neckerchief shall be examined for the defects listed in Table IV. The lot size shall be expressed in units of one neckerchief. The sample unit shall be one neckerchief.

TABLE IV. End item visual defects

Examine	Defect	Major	Minor
Material defects and workmanship damages	Holes, cut, tear, run, mend or slub	101	201
	Visible needle chew or needle hole		202
	Weakening defect in the material	102	
	Any thin area	103	
	Drop stitch		203
	Shade not comparable to standard or approved sample	104	
Seams and stitching	Wrong seam or stitch type	If it reduces wear and durability	204
	Seam twisted, puckered, or pleated affecting appearance		205
	Any line of stitching omitted or insecure		206
	Ends of overedge stitching not overlapping or overlapping less than ¾ inch	105	
	Ends of overedge stitching overlapping more than 1-¼ inches		207
	Two or more skipped stitches or one or more broken stitches	If it reduces wear and durability	208
	Thread breaks backstitched less than ½ inch	106	
	Loose tension resulting in loosely secured stitching	If it reduces wear and durability	209
	Stitches per inch less than minimum	107	
	Stitches per inch more than maximum		210
Label (marking)	Missing, incorrect, or illegible		211
	Not securely attached as specified, misplaced or stitching through printing		212

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TABLE IV. End item visual defects - Continued

Examine	Defect	Major	Minor
Label (marking) (cont'd)	Not applied on the back side of the cloth.		213
	Bar code not readable by scanner; human readable interpretation (HRI), omitted or illegible		214
	Bar code omitted	108	
	Bar code causes damage to the neckerchief	109	
	Bar code not visible on folded packaged item		215
Cleanliness	Spot or stain	110	
	Any marking exposed or visible on outside	111	
	Five or more thread ends not trimmed or removed		216
	Raw edges	112	
	Unpleasant odor		217
Dimensions and finished measurements	Not as specified.	113	
Component and Assembly	Any defect not specified in this table which affects, form, fit or function	114	

4.3.3 End item dimensional examination. The neckerchief sample shall be conditioned in accordance with ASTM D 1776 and then examined for conformance to the dimensions specified in 3.9. Any dimension not within the established tolerances as stated in 3.9 shall be classified as a defect. The lot size shall be expressed in units of one neckerchief. The sample unit shall be one neckerchief.

4.3.4 End item testing. The neckerchief shall be tested for the characteristics indicated in Table V. The methods of testing as specified in Table V shall be followed. All values specified represent an average value as determined by the applicable test method. All test reports shall contain the individual values utilized in expressing the final test data. The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

Lot size	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

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TABLE V. End item tests

Characteristics	Requirement paragraph	Test method
Fiber identification	3.4	ASTM D 276 or AATCC 20
Type of knit	3.4	Visual
Wales and courses	3.4	ASTM D 3887 1/
Bursting Strength	3.4	ASTM D 3787
Weight	3.4	ASTM D 3776 Opt C
Non-fibrous content	3.4	ASTM D 629 or AATCC 20A
pH	3.4	AATCC 81
Colorfastness to:		
Laundering (after 3 cycles)	3.5.3	AATCC-61, Opt 3A
Light (after 40hrs or 170 kilojoules)	3.5.3	AATCC-16, Opt 1 or 3
Crocking	3.5.3	AATCC 8
Perspiration	3.5.3	AATCC 15
Toxicity	3.6	4.4.1

1/ Fabric count only (#12)

4.4. Methods of inspection.

4.4.1 Toxicity test. When required (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of these studies indicate the neckerchief is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (See 2.3). If the toxicity requirement (see 3.6) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contactor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The neckerchief is intended for use by military personnel of the Department of Defense.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Class required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2).
d. When a first article is required (see 3.1, 4.2 and 6.3).
e. Toxicity requirements (see 3.6, and 4.4.1).
f. Bar Code Label requirements (3.8.1).
g. Conformance Inspection Quality Acceptance Limits (see 4.3).
h. Inspection Conditions (see 4.3.1).
i. Packaging (see 5.1).

6.3 First article. When a first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal. Standard samples are also available at DSCP through http://warfighter.dla.mil under tab “Vendor Info” then “Specifications/Pattern Request” under “Special Instructions” provide color shade, roll number and solicitation/contract number.

6.5 Knitted material. Knit material meeting the requirements of this specification was knit on a 22-cut machine using 28/1 carded yarns.

6.6 Subject term (key word) listing.
Neckwear
Scarf
Wrap

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians
Army – GL
Navy – NU
Air Force – 11

Preparing activity:
DLA - CT

Project No. 8440-2006-001

Review activities
Army – MD
Navy – MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at http://assist.daps.dla.mil/.

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